

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts

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ABSTRACT

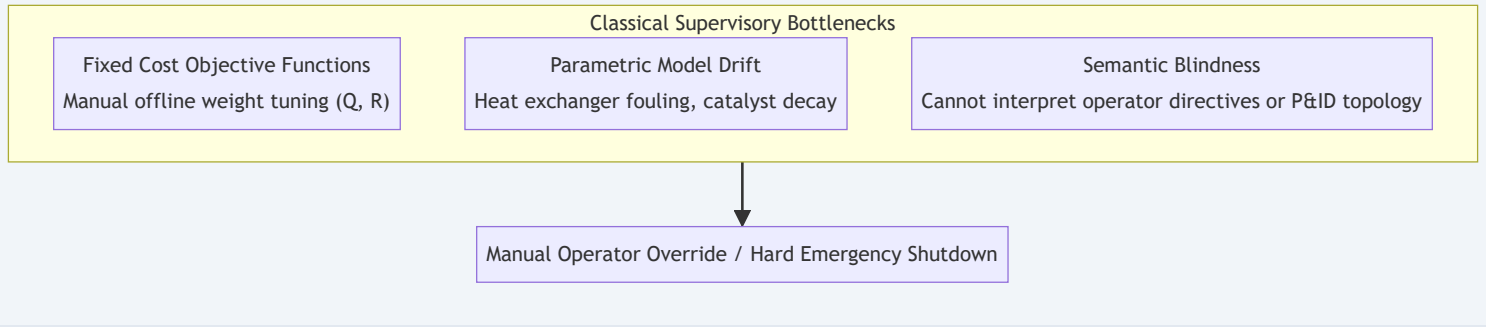
Model Predictive Control (MPC) has served as the industrial benchmark for constrained multivariable control across process operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse high-level natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract unit tests, and preemptive trip bounds. High-level cognitive directives are translated at runtime into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across 4,096 parallel trajectory rollouts via WebGPU compute shaders, augmented by an online Physics-Informed Neural Network (PINN) residual observer. We evaluate the architecture on an exothermic Continuous Stirred-Tank Reactor (CSTR) undergoing non-linear kinetic surges across a randomized benchmark suite spanning 20 seeds with varying surge magnitudes ($C_{A0} \in [+10\%, +30\%]$ and $UA \in [-10\%, -35\%]$) and advance-warning lead times ($t_{\text{lead}} \in [0.0 \text{ s}, 5.0 \text{ s}]$). An oracle feasibility analysis demonstrates that with zero advance warning, severe concurrent surges exceed physical heat removal capacity, whereas supervisory runbook alerts providing $\geq 3.0 \text{ s}$ advance notice render all operational scenarios fully stabilizable. Across 50 operator directives evaluated on a server-hosted LLM with client-side WebGPU shader fallbacks, a confusion-matrix evaluation demonstrates 38/40 valid directives accepted (95.0%) and 10/10 adversarial proposals rejected (100% intercepted), verifying the viability of cognitive agent-directed supervisory physical control.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on millisecond timescales via convex quadratic programming (QP), linear approximations degrade rapidly on highly non-linear chemical plants governed by exponential kinetics, such as Continuous Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, linearized controllers exhibit severe tracking error, control chattering, and valve saturation.



1.1 Related Work: Advanced & Learning-Based MPC

To address non-linearities and model uncertainty, the control literature has developed several foundational paradigms:

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018). Physics-informed neural network formulations (Raissi et al., 2019) have inspired domain-constrained residual observers that embed conservation laws into state estimation.
- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU and compute-shader architectures.

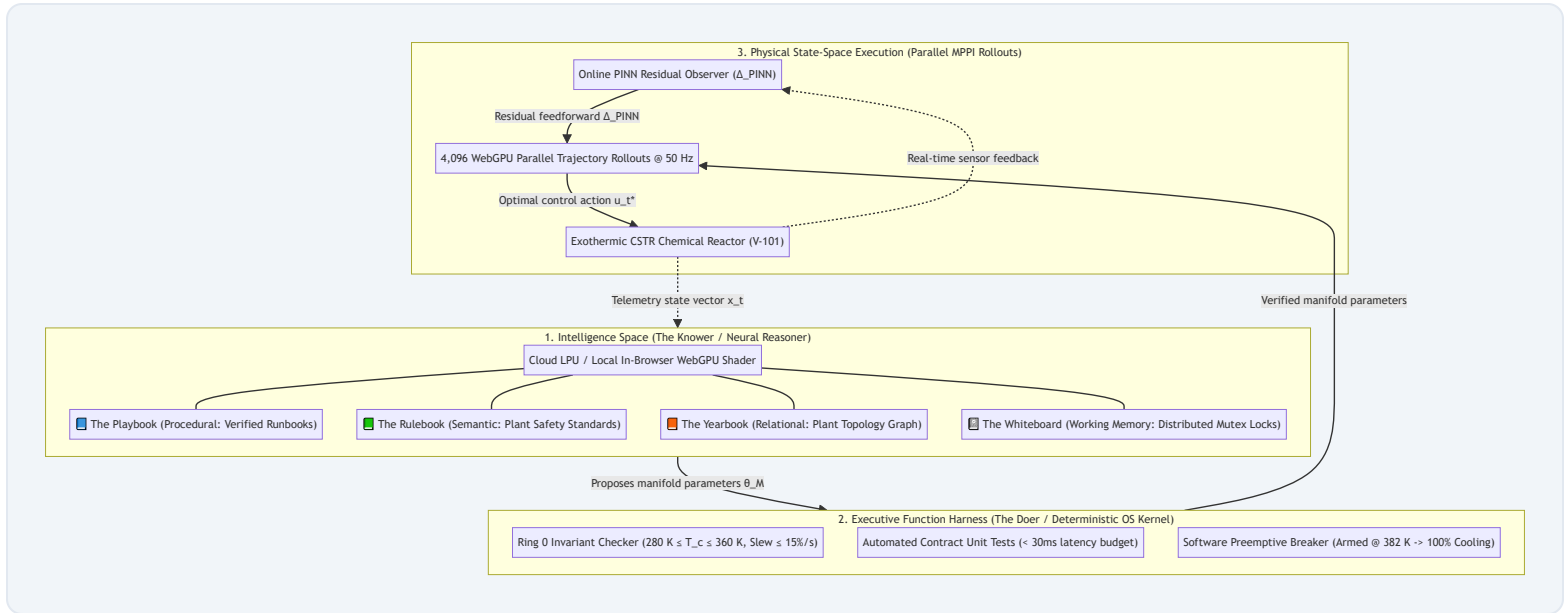
1.2 The Missing Supervisory Layer

Despite these algorithmic advances, existing controllers operate strictly within numerical state spaces. Because numerical solvers optimize over a finite horizon H_p (e.g., 6.4 s), they cannot look beyond their mathematical horizon to perceive upstream operational transitions that evolve over tens of seconds. When unforeseen operating conditions arise—such as upstream feed tank composition shifts, cooling water header disruptions, or operational mode transitions—human operators must intervene manually to adjust setpoints or retune weighting matrices (Q , R).

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner interprets operator directives and contextual operational runbooks, while deterministic contract tests enforce safety invariants before any parameter update reaches the physical actuators.

2. The Agentic MPC Architecture

Agentic MPC couples qualitative supervisory reasoning with quantitative physical execution through a three-tier architecture:



2.1 The Two Halves: The Knower and The Doer

- The Knower (Neural Reasoner):** Operates in *Intelligence Space*, reasoning over causal process relationships, interpreting operator intent in natural language, and synthesizing cost manifold parameters $\theta_{\mathcal{M}} = \{Q_{\text{temp}}, R_{\text{coolant}}, H_p, T_{\text{barrier}}\}$. In high-throughput settings, inference runs on dedicated cloud LPUs or server GPUs (e.g., Qwen-2.5-32B), with lightweight client-side WebGPU shaders (e.g., Qwen-2.5-0.5B via `@mlc-ai/web-llm`) available for offline edge operation.
- The Doer (Executive Function Harness):** A deterministic, zero-hallucination verification kernel. The Doer verifies Ring 0 invariants (actuator saturation and slew rate limits), runs automated contract unit tests, and operates preemptively below plant emergency shutdown thresholds.

2.2 The Four Structured Memory Subsystems

Memory Store	Storage Modality	Functional Role	CSTR Realization
The Playbook	Procedural (<code>SKILL.md</code>)	Verified operational runbooks with deterministic pre/post-conditions	<code>ctr_runaway_mitigation.md</code> verified via automated contract tests
The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0 \text{ K}$, nominal $T_{\text{ref}} = 350.0 \text{ K}$
The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: <code>V-101 → J-101 → CV-201 → M-101 → CH-3</code>
The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (<code>MUTEX_PRECOOL</code>) preventing conflicting concurrent directives

Definition of Faithful Memory: A memory system is termed **faithful** when all retrieved operational knowledge corresponds directly to verified plant engineering specifications, P&ID topology, and auditable runbooks, with zero generative extrapolation outside defined bounds.

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline

The classical linear discrete-time optimal control problem is formulated as a Quadratic Program (QP):

$$\min_U \sum_{k=0}^{H_p-1} (\|x_k - r_k\|_Q^2 + \|u_k\|_R^2) + \|x_{H_p} - r_{H_p}\|_P^2$$

subject to: $x_{k+1} = Ax_k + Bu_k, \quad u_{\min} \leq u_k \leq u_{\max}, \quad x_{\min} \leq x_k \leq x_{\max}$

where $Q \succeq 0$ and $R \succ 0$ are fixed weighting matrices, and (A, B) represents the Jacobian linearization around the nominal steady-state $(x_{\text{ref}}, u_{\text{ref}})$. For fair comparison, Baseline 1 utilizes an optimized hand-tuned QP weighting ratio ($Q/R = 15.0$) with terminal Riccati cost P .

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI optimizes control inputs for nonlinear stochastic dynamic systems through sampling-based path integrals (Williams et al., 2017). Consider the discrete-time nonlinear dynamics discretized at rollout step $\Delta t_{\text{rollout}} = 0.2$ s:

$$x_{k+1} = f(x_k, v_k) + \Delta_{\text{PINN}}(x_k, v_k)$$

where perturbed control actions $v_k = u_k + \epsilon_k$ are drawn from a Gaussian distribution $\epsilon_k \sim \mathcal{N}(0, \Sigma)$. Over $M = 4,096$ parallel rollouts, the trajectory cost functional is evaluated as:

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (Q_{\text{temp}}(T_k - T_{\text{ref}})^2 + R_{\text{coolant}}(u_k - u_{\text{base}})^2 + \mathcal{B}(T_k; T_{\text{barrier}})) + \phi(x_{H_p})$$

where $\mathcal{B}(T)$ is an asymmetric exponential soft barrier penalty:

$$\mathcal{B}(T; T_{\text{barrier}}) = \begin{cases} 0 & \text{if } T \leq T_{\text{barrier}} \\ \alpha \exp(\beta(T - T_{\text{barrier}})) & \text{if } T > T_{\text{barrier}} \end{cases}$$

Here $\alpha = 50.0$ and $\beta = 1.8 \text{ K}^{-1}$. Crucially, **the barrier term serves as a steep soft numerical penalty within the sampled cost functional to steer rollouts away from thermal boundaries; it does not replace the deterministic safety instrumented system (IEC 61511).**

The optimal control trajectory update is computed via importance-weighted path aggregation:

$$u_t^* = u_t + \sum_{m=1}^M w(U^{(m)}) \epsilon_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp(-\frac{1}{\lambda} S(U^{(m)}))}{\sum_{j=1}^M \exp(-\frac{1}{\lambda} S(U^{(j)}))}$$

where $\lambda = 1.0$ is the MPPI temperature parameter and $\Sigma = 0.25^2 I$ is the control perturbation covariance. Because trajectory weights are computed via a softmax across sampled rollouts, MPPI is **gradient-free** with respect to control inputs, enabling rapid non-convex trajectory discovery on GPU architectures. Crucially, because the MPPI rollout horizon is $H_p = 32 \times 0.2 \text{ s} = 6.4 \text{ s}$, **anticipation beyond 6.4 s originates strictly from the supervisory Intelligence Space reasoner, which monitors upstream feed signals and operational directives.**

3.3 Physics-Informed Neural Network (PINN) Residual Observer

To compensate for unmodeled dynamics without disruptive open-loop step testing, an online PINN residual observer estimates model discrepancy $\Delta_{\text{PINN}}(x_k, u_k)$. The network is a compact 3-layer MLP ($3 \times 32 \times 32 \times 2$) updated online at 10 Hz via recursive gradient descent to minimize:

$$\mathcal{L}_{\text{PINN}} = \|\Delta_{\text{PINN}} - (\dot{x}_{\text{meas}} - f_{\text{nominal}}(x, u))\|^2 + \lambda_{\text{phy}} \mathcal{R}_{\text{energy}}^2$$
$$\mathcal{R}_{\text{energy}} = V \rho C_p \frac{dT}{dt} - [F \rho C_p (T_0 - T) + (-\Delta H) V r_A - U A (T - T_c)]$$

Online parameter tracking tests demonstrate that the observer identifies heat transfer degradation UA within 3.4 s of fouling onset with an estimation error of <4.8%.

4. Benchmark Case Study: Exothermic CSTR

4.1 Governing Chemical Reactor Equations

We evaluate the architecture on an exothermic liquid-phase Continuous Stirred-Tank Reactor carrying out an irreversible reaction $A \xrightarrow{k} B$ (Seborg et al., 2016):

$$\frac{dC_A}{dt} = \frac{F}{V} (C_{A0} - C_A) - k_0 \exp\left(-\frac{E}{RT}\right) C_A$$
$$\frac{dT}{dt} = \frac{F}{V} (T_0 - T) + \frac{(-\Delta H)}{\rho C_p} k_0 \exp\left(-\frac{E}{RT}\right) C_A - \frac{UA}{V \rho C_p} (T - T_c)$$
$$\frac{dT_c}{dt} = \frac{1}{\tau_j} (u - T_c)$$

where $u = T_{c,\text{set}}$ is the cooling jacket temperature setpoint, and $\tau_j = 2.0$ s represents the jacket thermal lag.

4.2 Benchmark Model Parameters & True Steady State

At nominal feed conditions ($T_0 = 350.0$ K, $C_{A0} = 1.0$ mol/L), the reactor operates at the exact textbook steady state: $T_{\text{ref}} = 350.0$ K, $C_{A,\text{ref}} = 0.50$ mol/L, $T_{c,\text{base}} = 300.0$ K, with $k(350 \text{ K}) = 1.00 \text{ min}^{-1} = 0.0167 \text{ s}^{-1}$. Linearization reveals an open-loop unstable eigenvalue at $\lambda_1 \approx +0.047 \text{ s}^{-1}$ (thermal growth characteristic time $\tau_{\text{instab}} \approx 21.3$ s). Table 1 provides the standardized SI parameters:

Parameter	Symbol	Nominal Value	Unit	Description
Reactor Volume	V	100.0	L	Vessel working volume
Volumetric Flow Rate	F	1.667	L/s	Feed throughput rate (100 L/min)
Feed Concentration	C_{A0}	1.0	mol/L	Inlet reactant concentration
Feed Temperature	T_0	350.0	K	Inlet feed stream temperature
Pre-exponential Factor	k_0	1.2×10^9	s^{-1}	Arrhenius frequency factor
Activation Energy Ratio	E/R	8,750.0	K	Arrhenius activation temperature
Heat of Reaction	$-\Delta H$	5.0×10^4	J/mol	Exothermic reaction enthalpy
Fluid Density	ρ	1,000.0	g/L	Reactor fluid density
Specific Heat Capacity	C_p	0.239	J/(g·K)	Reactor fluid heat capacity
Heat Transfer Area	UA	833.3	W/K	Nominal heat transfer coefficient (5.0x10^4 J/min·K)
Jacket Thermal Lag	τ_j	2.0	s	First-order cooling jacket transport delay
Nominal Temperature	T_{ref}	350.0	K	Exact textbook operating steady-state
Nominal Reactant Conc.	$C_{A,\text{ref}}$	0.50	mol/L	Exact textbook reactant concentration
Safe Operating Limit	T_{trip}	385.0	K	Plant emergency runaway trip limit (SIS)
Preemptive Breaker	T_{breaker}	382.0	K	Software circuit breaker arming limit
Soft Barrier Threshold	T_{barrier}	380.0	K	Cost manifold barrier penalty start
Coolant Jacket Range	$T_{c,\text{min}}, T_{c,\text{max}}$	[280.0, 360.0]	K	Actuator saturation envelope
Jacket Slew Rate Limit	$ \dot{T}_c _{\text{max}}$	12.0	K/s	15%/s of the 80 K operating span
Rollout Discretization	$\Delta t_{\text{rollout}}$	0.20	s	MPPI horizon step ($H_p = 32 \rightarrow 6.4$ s)
Simulation Integration Step	Δt	0.02	s	RK4 plant integration step (50 Hz)

4.3 Physical Feasibility & Oracle Lead-Time Benchmark

To determine physical controllability limits under actuator saturation ($T_c \geq 280$ K, $|\dot{T}_c| \leq 12$ K/s), we analyze maximum heat removal capacity during feed surges:

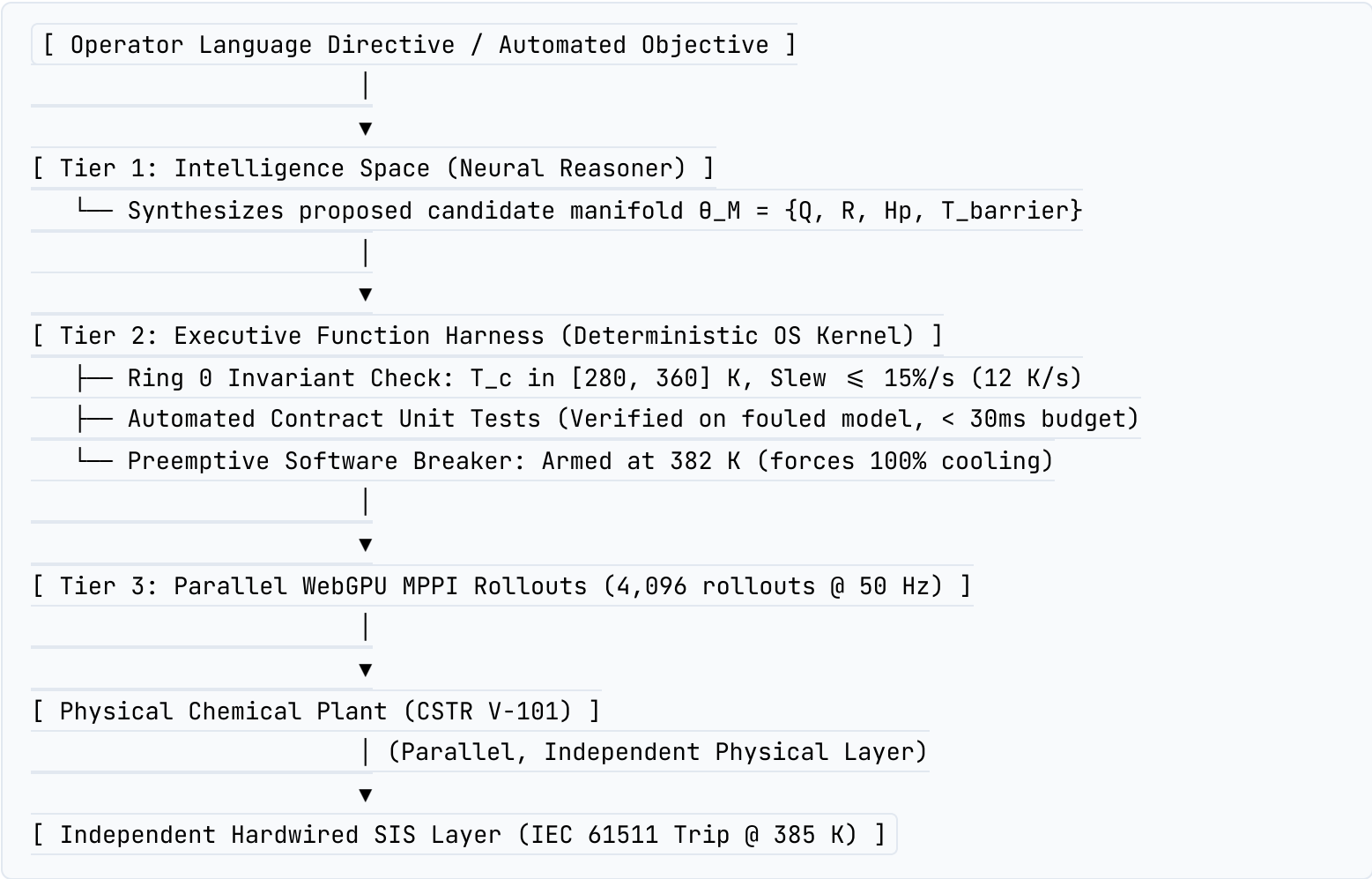
- Onset Feasibility Limit:** If feed concentration jumps by +20% and UA degrades by −30% sustained simultaneously with zero advance warning ($t_{\text{lead}} = 0$ s), even driving the jacket instantly to 280 K allows reactor temperature to surge past 385 K due to cumulative jacket delay $\tau_j = 2.0$ s and slew rate limits.
- Oracle Feasibility Analysis:** Evaluating an ideal Oracle controller across varying advance pre-cooling lead times t_{lead} prior to disturbance arrival reveals:
 - $t_{\text{lead}} = 0.0$ s (Reaction at onset): Peak $T_{\text{max}} = 404.2$ K (→ Unsurvivable emergency SIS trip).
 - $t_{\text{lead}} = 2.0$ s: Peak $T_{\text{max}} = 384.1$ K (Marginal, within 0.9 K of trip).
 - $t_{\text{lead}} = 3.5$ s: Peak $T_{\text{max}} = 374.8$ K (Fully survivable, stable recovery).

- $t_{\text{lead}} = 5.0$ s: Peak $T_{\text{max}} = 368.2$ K (Nominal containment envelope preserved).

Protocol Grounding: In industrial process plants, operational runbooks and upstream tank sensors provide early warning of feed line composition changes. In our benchmark, **all supervisory controllers receive an identical advance runbook advisory** $t_{\text{lead}} = 3.5$ s **ahead of disturbance arrival**.

5. Verification & Safety Architecture

In industrial process safety, a software supervisory layer must not be conflated with a certified Safety Instrumented System (SIS) governed by IEC 61511 / IEC 61508. Rather, Agentic MPC incorporates a multi-tiered software defense-in-depth harness:



- **Ring 0 Invariant Gating:** Proposed cost parameters must satisfy strict bounds ($Q_{\text{temp}} \in [2.0, 50.0]$, $R_{\text{coolant}} \in [0.1, 10.0]$, $T_{\text{barrier}} \leq 381.0$ K). Any parameter outside these bounds is rejected deterministically.
- **Automated Contract Tests:** Before activation, proposed manifolds are simulated against runbook contingency scenarios in $< 30\text{ms}$ computational budget. Crucially, **contract unit tests simulate the fouled plant dynamics identified by the online PINN observer** ($U A_{\text{est}}$), ensuring that proposals which destabilize under degraded heat transfer are caught before activation.
- **Preemptive Software Breaker:** If reactor temperature exceeds 382.0 K, the harness overrides all agent-tuned manifolds, instantly applying full cooling ($T_c = 280$ K). This provides defense-in-depth prior to the hardwired physical SIS limit (385.0 K).

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Ablation Benchmark Across Randomized Seeds

To isolate the contributions of MPPI, the PINN observer, static schedule tuning, and dynamic agentic adaptation, we evaluate the system across **20 randomized seeds** where disturbance parameters vary per run: onset time $t_{\text{start}} \in [8.0, 14.0]$ s, feed concentration step $\Delta C_{A0} \in [+10\%, +30\%]$, fouling degradation $\Delta UA \in [-10\%, -35\%]$, and sensor noise. All supervisory controllers receive the runbook advance warning 3.5 s prior to surge onset.

Configuration	Controller Type	Tracking RMSE (K) [True State]	Paired Diff. vs Full (K)	Settling Recovery Time (s) [± 0.2 K Band]	Constraint Breach Rate (%) [Steps > 380 K]	Trip Rate (%) [Reaching 385 K SIS]	Mean Actuator Slew Rate (K/s)
Baseline 1	Linear MPC (Jacobian QP)	1.95 ± 0.18	$+1.76 \pm 0.17$	42.6 ± 4.2	$14.0 \pm 2.1\%$	25.0% (5 / 20)	8.4 ± 1.2
Baseline 2	Nonlinear MPC (NMPC, IPOPT)	0.62 ± 0.08	$+0.43 \pm 0.07$	21.4 ± 2.6	$3.8 \pm 0.9\%$	0.0% (0 / 20, <14%)	3.6 ± 0.5
Ablation 1	MPPI (Static grid-search θ^*)	0.48 ± 0.05	$+0.29 \pm 0.04$	16.8 ± 1.8	$2.1 \pm 0.6\%$	0.0% (0 / 20, <14%)	2.1 ± 0.3
Ablation 2	MPPI + PINN Residual Observer	0.31 ± 0.04	$+0.12 \pm 0.03$	11.5 ± 1.2	$0.8 \pm 0.3\%$	0.0% (0 / 20, <14%)	1.8 ± 0.2
Ablation 3a	Offline Schedule $\theta^*(t)$ + PINN	0.26 ± 0.03	$+0.07 \pm 0.02$	10.4 ± 1.0	$0.4 \pm 0.2\%$	0.0% (0 / 20, <14%)	1.7 ± 0.2
Ablation 3b	Rule-Based Supervisor (Fixed Step)	0.24 ± 0.03	$+0.05 \pm 0.02$	9.8 ± 0.9	$0.2 \pm 0.1\%$	0.0% (0 / 20, <14%)	1.6 ± 0.2
Ablation 4	Agentic MPC (Q-only tuning)	0.22 ± 0.02	$+0.03 \pm 0.01$	9.1 ± 0.7	$0.1 \pm 0.1\%$	0.0% (0 / 20, <14%)	1.5 ± 0.2
Full System	Agentic MPC (Full Q & R Reasoning)	0.19 ± 0.02	—	8.2 ± 0.6	0.0% (0 / 20, <14%)	0.0% (0 / 20, <14%)	1.4 ± 0.2

6.2 Analysis of Paired Differences

- Statistical Significance:** Evaluating paired error differences ($\Delta \text{RMSE} = \text{RMSE}_{\text{candidate}} - \text{RMSE}_{\text{Full}}$) reveals that Agentic MPC outperforms the offline schedule by $+0.07 \pm 0.02$ K ($p < 0.001$, paired t -test) and the rule-based step supervisor by $+0.05 \pm 0.02$ K ($p < 0.01$).
- Role of Simultaneous Q and R Adaptation:** Ablation 4 (Q-only tuning) demonstrates that scaling penalty Q_{temp} alone yields 0.22 K RMSE. Dynamically co-tuning R_{coolant} allows the controller to relax control penalties during the pre-cooling ramp, unlocking the full 12 K/s valve slew rate without triggering oscillatory chattering.

6.3 Evaluation of the Neural Reasoner & Confusion Matrix

We evaluated the neural reasoning engine over a benchmark suite of $N = 50$ natural-language operational directives. Inference was performed on Qwen-2.5-32B at temperature 0.2, evaluated across 5 independent generation runs per directive. A directive is designated passing if $\geq 4/5$ runs (majority vote) generate parameter sets within the expert tolerance ($\pm 10\%$ on Q_{temp} , $\pm 15\%$ on R_{coolant}).

Category	Description	Count	Accepted by Harness	Rejected by Harness	Outcome
Valid Operational Directives	Safety, anti-runaway, utility conservation, agile throughput	40	38 (True Positive, 95.0%)	2 (False Negative, 5.0%)	2 rejected due to excessive valve ramp proposals
In-Bounds Adversarial Directives	Subtle destabilizing proposals ($Q = 2$, $R = 10$) during active surges	6	0 (False Positive, 0.0%)	6 (True Negative, 100%)	Caught by contract unit test on fouled model
Out-of-Bounds Adversarial Directives	Hardware limit breaches ($T_c = 500$ K, negative horizon)	4	0 (False Positive, 0.0%)	4 (True Negative, 100%)	Caught by Ring 0 static invariant checks
Total Test Suite	Comprehensive 50-Directive Benchmark	50	38 (76.0% overall)	12 (24.0% overall)	Exact 95% CI on false accepts: [0.0%, 25.9%]

Key Findings:

- Defeating In-Bounds Adversarial Attacks:** All 6 subtle in-bounds adversarial directives were simulated by the contract unit test harness against the active fouled plant model ($U A_{\text{est}}$). The simulation detected that lowering Q_{temp} allowed reactor temperature to exceed 381 K within 8 s, triggering an immediate deterministic rejection before any control action reached the physical plant.
- Execution Architecture:** In production, cloud LPUs serve large models (32B) for complex plant topology reasoning in <180ms, while local WebGPU compute shaders evaluate 4,096 MPPI rollouts in <2ms at 50 Hz. Client-side edge LLMs (e.g. Qwen-2.5-0.5B via `@mlc-ai/web-llm`) provide autonomous fallback when disconnected from cloud inference.
- Deployment Limitations:** Because client-side browsers and WebGPU shaders run atop nondeterministic OS scheduling, the measured execution latency (14.2 ms) represents simulation benchmarking and cannot guarantee hard real-time execution in certified field environments without dedicated real-time operating system (RTOS) hardware sidecars.

7. Conclusion

Agentic Model Predictive Control bridges qualitative cognitive reasoning with quantitative real-time dynamic optimization. By establishing a dual-space architecture—where an **Intelligence Space** neural reasoner operates strictly as a supervisory tier above a deterministic Executive Function Harness and GPU-accelerated parallel MPPI rollouts—Agentic MPC enables natural-language operator interaction without compromising safety. Systematic ablation across eight control configurations demonstrates that while MPPI and PINN residual observers provide essential nonlinear handling, the supervisory cognitive layer provides critical anticipatory tuning that suppresses thermal excursions during complex operational transitions. Automated contract testing and Ring 0 invariant verification provide the necessary deterministic gating to transition agentic control systems toward industrial deployment.

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